Capote 2026-05-04 Fit Chain: Step-by-Step Benchmark Report

This report documents the implementation of the new TNC fit chain proposed by R. Capote in his 2026-05-04 emails *"NEW TNC FIT steps"* and *"Revision/additions to Axton Total cross sections experimental data"*. Each section corresponds to one of the seven steps (0–6) in the chain, showing how the iterative Generalized Least Squares (GLS) fit results evolve as data are added, deleted, reclassified, or modified.

All fits use the microscopic-only data subset (Maxwellian-derived functions excluded) on the Axton 1986 database, starting from the no-Lounsbury baseline that reproduces Axton Table 8 (see `report_axton1986_reproduction.md`).

Quantities with shifts exceeding 0.1% relative to the previous step are highlighted in **bold**. The Westcott-folded quantities FA, FF, F1ETA, F2ETA, F3ETA are computed as derived outputs at each step per Roberto's request (K1 = F3ETA).

Overview

#	Step	N meas	N free	DOF	χ^2	Converged
0	Axton T8 baseline (no Lounsbury)	102	30	72	55.13	Yes
1	+ Lounsbury alpha (21–23)	105	30	75	60.49	Yes
2	+ Adamchuk 1988 alpha(^{235}U)	106	30	76	60.56	Yes
3	+ Wallner 2014 U-235 capture	108	30	78	62.85	Yes
4	Sears SCA list + Pu-241 estimate; delete Lemmel SCA	118	30	88	74.68	Yes
5	+ transmission additions; 3 SCR→SCA reclassifications	128	30	98	84.21	Yes
6	All ABS+SCR → ABS+SCA; drop Lemmel SCA-SCR; SCR removed	124	26	98	85.18	Yes

Roberto's targeted comparison: Step 5 vs Step 6

Roberto's email asks to *"compare steps 5 and 6 — hopefully one of them will be the final fit"*. The two candidate fits agree to better than 0.2% on every primary and derived quantity. Step 5 retains 4 SCR fit variables constrained by Lemmel SCA–SCR datapoints; step 6 removes them per Roberto's parenthetical *"reducing 4 fitting variables SCR"*. Detailed numbers are in the per-step sections below; the head-to-head comparison appears in the dedicated section after step 6.

Step 0: Axton Table 8 Baseline (No Lounsbury)

This is the reference point: the microscopic GLS fit reproducing Axton's Table 8 results, with Lounsbury alpha measurements (Nos. 21–23) excluded. The detailed reproducibility analysis and the rationale for excluding Lounsbury are documented in `report_axton1986_reproduction.md` (Section 4); briefly, Lounsbury alpha measurements carry a "Z" systematic code (*Lounsbury Temperature Uncertainty*) that makes them effectively Maxwellian/integral, and Axton excluded them in his Table 6/Table 8 microscopic fit.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3206	5.681	–
SCA 35	16.2235	7.385	–
SCA 39	7.9001	12.380	–
SCA 41	11.9720	21.596	–
SCR 33	11.1483	8.161	–
SCR 35	14.5068	9.455	–
SCR 39	6.8449	24.890	–
SCR 41	10.8234	35.520	–
ABS 33	575.2413	0.321	–
ABS 35	681.2060	0.249	–
ABS 39	1018.7914	0.362	–
ABS 41	1382.2612	1.079	–
FIS 33	533.1364	0.460	–
FIS 35	585.0334	0.277	–
FIS 39	748.5245	0.340	–
FIS 41	1020.3307	1.127	–
CA 33	42.1049	4.228	–
CA 35	96.1726	1.811	–
CA 39	270.2669	1.181	–
CA 41	361.9304	1.729	–
CAP 34	95.9000	2.086	–
NUB 33	2.4860	0.217	–

Quantity	Value	Unc (%)	Δ vs prev (%)
NUB 35	2.4264	0.188	–
NUB 39	2.8794	0.207	–
NUB 41	2.9406	0.220	–
NUB 52	3.7649	0.132	–
ETA 33	2.3040	0.269	–
ETA 35	2.0838	0.255	–
ETA 39	2.1156	0.310	–
ETA 41	2.1706	0.399	–
ALPHA 33	0.0790	4.545	–
ALPHA 35	0.1644	1.962	–
ALPHA 39	0.3611	1.289	–
ALPHA 41	0.3547	1.697	–
WGA 33	0.9988	0.120	–
WGA 35	0.9787	0.092	–
WGA 39	1.0767	0.269	–
WGA 41	1.0444	0.191	–
WGF 33	0.9965	0.201	–
WGF 35	0.9778	0.092	–
WGF 39	1.0562	0.275	–
WGF 41	1.0452	0.670	–
FA 33	574.5740	0.343	–
FA 35	666.6963	0.265	–
FA 39	1096.9327	0.451	–
FA 41	1443.6336	1.096	–
FF 33	531.2491	0.502	–
FF 35	572.0457	0.292	–
FF 39	790.5916	0.437	–
FF 41	1066.4497	1.312	–
F1ETA 33	2.2985	0.357	–
F1ETA 35	2.0819	0.286	–
F1ETA 39	2.0753	0.494	–
F1ETA 41	2.1723	0.803	–
F2ETA 33	1320.6836	0.463	–
F2ETA 35	1387.9982	0.303	–
F2ETA 39	2276.4588	0.453	–
F2ETA 41	3135.9879	1.304	–
F3ETA 33	746.1097	0.680	–
F3ETA 35	721.3019	0.507	–
F3ETA 39	1179.5261	0.803	–
F3ETA 41	1692.3543	1.773	–

Step 1: + Lounsbury Alpha (Nos. 21–23)

Step 1 adds the three Lounsbury 1970 alpha measurements back into the microscopic fit:

- No.21 LOUNSBURY (CA 33)/FIS 33 = 0.0861, σ = 2.24%
- No.22 LOUNSBURY (CA 35)/FIS 35 = 0.1697, σ = 1.71%
- No.23 LOUNSBURY (CA 39)/FIS 39 = 0.3555, σ = 1.60%

This is the largest single change in the entire chain. Lounsbury's α(^{233}U) = 0.0861 is far above Axton's converged value of 0.0787, and pulling the system toward this value drives a large redistribution of ABS(^{233}U) between capture and fission — ALPHA(^{233}U) shifts by +7.2%, CA(^{233}U) by +6.6%, and FIS(^{233}U) drops by 0.5%. F3ETA(^{233}U) drops by 0.56%.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3166	5.681	-0.032
SCA 35	**15.8847**	7.270	**-2.088**
SCA 39	**7.9230**	12.364	**+0.291**
SCA 41	11.9618	21.596	-0.085
SCR 33	11.1419	8.171	-0.057
SCR 35	**14.0797**	9.527	**-2.944**
SCR 39	**6.9469**	23.295	**+1.490**
SCR 41	**10.8049**	35.707	**-0.170**
ABS 33	575.2692	0.321	+0.005
ABS 35	**682.0852**	0.233	**+0.129**
ABS 39	1018.2345	0.340	-0.055
ABS 41	1382.7102	1.079	+0.032

Quantity	Value	Unc (%)	Δ vs prev (%)
FIS 33	**530.3843**	0.355	**-0.516**
FIS 35	**584.0124**	0.244	**-0.175**
FIS 39	748.1165	0.321	-0.055
FIS 41	**1018.4694**	1.119	**-0.182**
CA 33	**44.8849**	1.837	**+6.603**
CA 35	**98.0728**	1.152	**+1.976**
CA 39	270.1180	0.913	-0.055
CA 41	**364.2408**	1.663	**+0.638**
CAP 34	95.9000	2.086	+0.000
NUB 33	**2.4910**	0.184	**+0.203**
NUB 35	**2.4294**	0.172	**+0.123**
NUB 39	2.8814	0.192	+0.067
NUB 41	2.9435	0.209	+0.099
NUB 52	3.7664	0.129	+0.039
ETA 33	**2.2967**	0.199	**-0.319**
ETA 35	**2.0801**	0.211	**-0.180**
ETA 39	2.1170	0.267	+0.067
ETA 41	**2.1681**	0.391	**-0.116**
ALPHA 33	**0.0846**	1.960	**+7.156**
ALPHA 35	**0.1679**	1.235	**+2.154**
ALPHA 39	0.3611	0.965	-0.001
ALPHA 41	**0.3576**	1.585	**+0.822**
WGA 33	0.9988	0.120	+0.000
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	574.6019	0.343	+0.005
FA 35	**667.5568**	0.250	**+0.129**
FA 39	1096.3331	0.433	-0.055
FA 41	1444.1025	1.096	+0.032
FF 33	**528.5068**	0.408	**-0.516**
FF 35	**571.0473**	0.261	**-0.175**
FF 39	790.1607	0.423	-0.055
FF 41	**1064.5042**	1.305	**-0.182**
F1ETA 33	**2.2912**	0.307	**-0.319**
F1ETA 35	**2.0782**	0.248	**-0.180**
F1ETA 39	2.0767	0.468	+0.067
F1ETA 41	**2.1698**	0.799	**-0.116**
F2ETA 33	**1316.5317**	0.428	**-0.314**
F2ETA 35	1387.2857	0.296	-0.051
F2ETA 39	2276.7426	0.451	+0.012
F2ETA 41	3133.3762	1.302	-0.083
F3ETA 33	**741.9298**	0.603	**-0.560**
F3ETA 35	**719.7289**	0.470	**-0.218**
F3ETA 39	1180.4095	0.781	+0.075
F3ETA 41	**1689.2737**	1.769	**-0.182**

Step 2: + Adamchuk 1988 Alpha(^{235}U)

Step 2 adds one new measurement (No.168):

- ADAMCHUK (CA 35)/FIS 35 = 0.169, σ = 2.1%, At. En. (R) 65, 434 (1988) (GMA NS=8026)

Adamchuk's α(^{235}U) = 0.169 is highly consistent with the step-1 fitted value (0.168), so the step is essentially a no-op for χ^2: only +0.07 from one added measurement, and shifts in U-235 quantities are at the 0.1–0.2% level. All other nuclides are unchanged.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3166	5.681	-0.000
SCA 35	**15.8609**	7.245	**-0.150**
SCA 39	7.9233	12.364	+0.003
SCA 41	11.9618	21.596	-0.000
SCR 33	11.1419	8.171	-0.000
SCR 35	**14.0487**	9.509	**-0.220**

Quantity	Value	Unc (%)	Δ vs prev (%)
SCR 39	6.9479	23.281	+0.014
SCR 41	10.8049	35.708	-0.000
ABS 33	575.2694	0.321	+0.000
ABS 35	682.1500	0.230	+0.010
ABS 39	1018.2347	0.340	+0.000
ABS 41	1382.7113	1.079	+0.000
FIS 33	530.3690	0.355	-0.003
FIS 35	583.9317	0.239	-0.014
FIS 39	748.0666	0.320	-0.007
FIS 41	1018.3865	1.119	-0.008
CA 33	44.9004	1.832	+0.035
CA 35	**98.2183**	0.999	**+0.148**
CA 39	270.1681	0.910	+0.019
CA 41	364.3248	1.660	+0.023
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4911	0.184	+0.002
NUB 35	2.4295	0.170	+0.006
NUB 39	2.8815	0.191	+0.003
NUB 41	2.9436	0.208	+0.004
NUB 52	3.7664	0.129	+0.002
ETA 33	2.2967	0.199	-0.001
ETA 35	2.0797	0.200	-0.017
ETA 39	2.1169	0.266	-0.003
ETA 41	2.1680	0.390	-0.005
ALPHA 33	0.0847	1.955	+0.037
ALPHA 35	**0.1682**	1.064	**+0.162**
ALPHA 39	0.3612	0.960	+0.025
ALPHA 41	0.3577	1.580	+0.031
WGA 33	0.9988	0.120	+0.000
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	574.6021	0.343	+0.000
FA 35	667.6202	0.248	+0.010
FA 39	1096.3333	0.433	+0.000
FA 41	1444.1037	1.096	+0.000
FF 33	528.4915	0.408	-0.003
FF 35	570.9684	0.256	-0.014
FF 39	790.1079	0.422	-0.007
FF 41	1064.4175	1.304	-0.008
F1ETA 33	2.2912	0.307	-0.001
F1ETA 35	2.0778	0.239	-0.017
F1ETA 39	2.0766	0.468	-0.003
F1ETA 41	2.1697	0.798	-0.005
F2ETA 33	1316.5210	0.428	-0.001
F2ETA 35	1387.1801	0.295	-0.008
F2ETA 39	2276.6671	0.451	-0.003
F2ETA 41	3133.2351	1.302	-0.005
F3ETA 33	741.9189	0.603	-0.001
F3ETA 35	719.5599	0.461	-0.023
F3ETA 39	1180.3338	0.781	-0.006
F3ETA 41	1689.1314	1.769	-0.008

Step 3: + Wallner 2014 U-235 Capture

Step 3 adds two Wallner et al. 2014 thermal capture measurements (PRL 112, 192501, 2014):

- No.169 WALLNER CA 35 = 99.5 b, σ = 2.0%
- No.170 WALLNER CA 35 = 101.5 b, σ = 2.0%

Both Wallner values lie above the step-2 fit (CA 35 ≈ 98.2 b). The fit accommodates the pull with $\chi^2 = +2.3$ over 2 added measurements; CA(^{235}U) shifts +0.76% to 98.97 b. Through the absorption channel correlations, SCA(^{235}U) drops 1.3% and SCR(^{235}U) drops 1.9%. F3ETA(^{235}U) shifts only 0.08%, so the K1 of interest is essentially preserved.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	12.3165	5.681	-0.001
SCA 35	**15.6565**	7.185	**-1.288**
SCA 39	7.9214	12.364	-0.024
SCA 41	11.9602	21.596	-0.013
SCR 33	11.1417	8.172	-0.002
SCR 35	**13.7764**	9.597	**-1.938**
SCR 39	**6.9407**	23.380	**-0.103**
SCR 41	10.8020	35.738	-0.027
ABS 33	575.2702	0.321	+0.000
ABS 35	682.7209	0.220	+0.084
ABS 39	1018.3508	0.340	+0.011
ABS 41	1382.7815	1.079	+0.005
FIS 33	530.2947	0.355	-0.014
FIS 35	583.7524	0.237	-0.031
FIS 39	747.9269	0.320	-0.019
FIS 41	1018.0396	1.119	-0.034
CA 33	**44.9755**	1.826	**+0.167**
CA 35	**98.9685**	0.814	**+0.764**
CA 39	270.4239	0.907	+0.095
CA 41	**364.7420**	1.657	**+0.114**
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4913	0.184	+0.010
NUB 35	2.4303	0.169	+0.030
NUB 39	2.8819	0.191	+0.016
NUB 41	2.9441	0.208	+0.018
NUB 52	3.7667	0.129	+0.007
ETA 33	2.2966	0.199	-0.004
ETA 35	2.0780	0.190	-0.084
ETA 39	2.1166	0.266	-0.014
ETA 41	2.1675	0.390	-0.022
ALPHA 33	**0.0848**	1.949	**+0.181**
ALPHA 35	**0.1695**	0.879	**+0.795**
ALPHA 39	**0.3616**	0.956	**+0.113**
ALPHA 41	**0.3583**	1.574	**+0.149**
WGA 33	0.9988	0.120	+0.000
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	574.6028	0.343	+0.000
FA 35	668.1790	0.239	+0.084
FA 39	1096.4583	0.433	+0.011
FA 41	1444.1770	1.096	+0.005
FF 33	528.4174	0.408	-0.014
FF 35	570.7931	0.255	-0.031
FF 39	789.9604	0.422	-0.019
FF 41	1064.0549	1.304	-0.034
F1ETA 33	2.2911	0.307	-0.004
F1ETA 35	2.0761	0.230	-0.084
F1ETA 39	2.0763	0.468	-0.014
F1ETA 41	2.1692	0.798	-0.022
F2ETA 33	1316.4696	0.428	-0.004
F2ETA 35	1387.1758	0.295	-0.000
F2ETA 39	2276.6053	0.451	-0.003
F2ETA 41	3132.7161	1.302	-0.017
F3ETA 33	741.8668	0.603	-0.007
F3ETA 35	718.9968	0.458	-0.078
F3ETA 39	1180.1470	0.781	-0.016
F3ETA 41	1688.5391	1.769	-0.035

Step 4: Sears SCA List + Pu-241 Estimate; Delete Lemmel SCA

Step 4 implements the most extensive change in the chain. It replaces the four Lemmel direct SCA measurements with a curated compilation drawn from Sears (1992), the Divadeenam compilation, EXFOR replacements where Roberto annotated them, and Pronyaev's potential-scattering estimate for Pu-241.

The Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95) are *kept* per Roberto's instruction *"leave SCA-SCR"*; only the four direct

Lemmel SCA datapoints are deleted.

Datapoints deleted

No.	Author	Reference	Quantity	Value	Note
88	Lemmel	1982	SCA(^{233}U)	12.306	replaced by step-4 SCA list
90	Lemmel	1982	SCA(^{235}U)	16.508	replaced by step-4 SCA list
92	Lemmel	1982	SCA(^{239}Pu)	8.004	replaced by step-4 SCA list
94	Lemmel	1982	SCA(^{241}Pu)	12.006	replaced by Pronyaev 12.0(0.8)

Datapoints added

No.	Author / year	Reference	Type	Value (b)	σ (b)	Appendix
171	Atoyi	Acta Cryst. 20, 587 (1966)	SCA(^{235}U)	12.3	0.5	A2
172	Willis	PRS A274, 122 (1963)	SCA(^{235}U)	12.8	1	A3
173	Kaiser	Physica B 136, 134 (1986)	SCA(^{235}U)	14	0.2	
174	Arif	PR A35, 2810 (1987)	SCA(^{235}U)	14	0.2	
175	Ceulemans	Divadeenam compilation 1970	SCA(^{235}U)	14.6	0.5	A4
176	Bayvas	Divadeenam compilation 1980	SCA(^{235}U)	14.3	1.5	
177	Foote	Divadeenam compilation 1958	SCA(^{235}U)	15	1	
178	Oleksa	Phys Rev 109, 1645 (1958)	SCA(^{233}U)	13.4	0.5	A5
179	Green	JNM 51, 281 (1974)	SCA(^{233}U)	12.6	0.7	A6
180	Vertebnyi	Divadeenam compilation 1974	SCA(^{233}U)	13.2	0.6	
181	Green	JNM 34, 281 (1970)	SCA(^{239}Pu)	7.7	0.6	
182	Atoyi	Acta Cryst. 20, 587 (1966)	SCA(^{239}Pu)	7.3	0.7	
183	Roof	Acta Cryst. 15, 351 (1962)	SCA(^{239}Pu)	7.3	0.7	
184	Pronyaev	Internal analysis 2026-04-17	SCA(^{241}Pu)	12	0.8	A7

Impact The U-235 SCA channel is the most affected: SCA(^{235}U) drops from 15.66 b to 13.96 b — a 10.9% shift — driven by the Kaiser/Arif interferometry consensus around 14.0 b replacing the 16-ish Lemmel value. SCA uncertainties collapse dramatically (e.g. SCA 35: 7.19% ⟶ 0.92%; SCA 41: 21.6% ⟶ 6.7% via Pronyaev's pin), reflecting the data richness of the new compilation. ABS, FIS, CA, and the F-quantities shift by ≤0.16% — the K1 = F3ETA is essentially insensitive to the SCA reshaping because F3ETA = NUB·FF − FA does not directly depend on SCA. χ^2 rises by 11.9 from a net +10 measurements (~1.2/measurement, indicating moderate but absorbable tension).

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	**13.1687**	2.559	**+6.919**
SCA 35	**13.9550**	0.922	**-10.868**
SCA 39	**7.4588**	5.107	**-5.840**
SCA 41	**11.9950**	6.665	**+0.291**
SCR 33	**11.9926**	5.624	**+7.637**
SCR 35	**12.1748**	6.230	**-11.626**
SCR 39	**6.4499**	21.795	**-7.071**
SCR 41	**10.8309**	27.719	**+0.268**
ABS 33	**574.4310**	0.304	**-0.146**
ABS 35	**683.8756**	0.190	**+0.169**
ABS 39	1018.9603	0.336	+0.060
ABS 41	1383.0480	1.070	+0.019
FIS 33	**529.5169**	0.340	**-0.147**
FIS 35	**584.6560**	0.214	**+0.155**
FIS 39	748.4082	0.317	+0.064
FIS 41	1018.2307	1.111	+0.019
CA 33	**44.9141**	1.823	**-0.136**
CA 35	**99.2196**	0.796	**+0.254**
CA 39	270.5521	0.904	+0.047
CA 41	364.8174	1.650	+0.021
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4914	0.184	+0.001
NUB 35	2.4303	0.169	+0.004
NUB 39	2.8819	0.191	+0.000
NUB 41	2.9442	0.208	+0.001
NUB 52	3.7667	0.129	+0.001
ETA 33	2.2966	0.199	+0.000

Quantity	Value	Unc (%)	Δ vs prev (%)
ETA 35	2.0777	0.190	-0.011
ETA 39	2.1167	0.266	+0.005
ETA 41	2.1676	0.390	+0.001
ALPHA 33	0.0848	1.949	+0.010
ALPHA 35	0.1697	0.876	+0.099
ALPHA 39	0.3615	0.954	-0.017
ALPHA 41	0.3583	1.574	+0.002
WGA 33	0.9988	0.120	+0.000
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	**573.7647**	0.327	**-0.146**
FA 35	**669.3091**	0.211	**+0.169**
FA 39	1097.1146	0.431	+0.060
FA 41	1444.4553	1.087	+0.019
FF 33	**527.6424**	0.395	**-0.147**
FF 35	**571.6766**	0.233	**+0.155**
FF 39	790.4688	0.420	+0.064
FF 41	1064.2547	1.298	+0.019
F1ETA 33	2.2911	0.307	+0.000
F1ETA 35	2.0758	0.230	-0.011
F1ETA 39	2.0764	0.468	+0.005
F1ETA 41	2.1692	0.798	+0.001
F2ETA 33	**1314.5515**	0.415	**-0.146**
F2ETA 35	**1389.3732**	0.275	**+0.158**
F2ETA 39	2278.0813	0.449	+0.065
F2ETA 41	3133.3398	1.295	+0.020
F3ETA 33	**740.7869**	0.594	**-0.146**
F3ETA 35	**720.0641**	0.447	**+0.148**
F3ETA 39	1180.9667	0.780	+0.069
F3ETA 41	1688.8844	1.764	+0.020

Step 5: + Transmission Entries; 3 ABS+SCR → ABS+SCA Reclassifications

Step 5 implements Roberto's transmission-data revision from his "Revision/additions to Axton Total cross sections experimental data" email of 2026-05-04. Three existing transmission entries are reclassified, ten new entries are added, and two would-be additions (Spencer 1987) are recognised as already in the database.

Datapoints reclassified (ABS+SCR → ABS+SCA)

No.	Author	Year	Nuclide	Change	Appendix
68	Safford	1959	U-235	ABS+SCR → ABS+SCA	
78	Nikitin	1955	U-233	ABS+SCR → ABS+SCA	A8
83	Safford	1960	U-233	ABS+SCR → ABS+SCA	A9

Datapoints added

No.	Author / year	Reference	Type	Value (b)	σ (b)	Appendix
185	Antonov	1986	ABS+SCA(^{235}U)	695	15	A10
186	Gerasimov	1962	ABS+SCR(^{235}U)	685	9	A11
187	Leonard	1954	ABS+SCR(^{235}U)	702	14	A12
188	Pschenichnyi	1978	ABS+SCR(^{233}U)	587.9	3	A13
189	Harvey-1	1979	ABS+SCR(^{233}U)	585	1.4	A14
190	Harvey-2	1979	ABS+SCR(^{233}U)	592	6	A14
191	Schwartz	1958	ABS+SCR(^{241}Pu)	1410	80	
192	Anderson	1945	ABS+SCR(^{239}Pu)	1045	25	
193	Palevsky	1955	ABS+SCR(^{239}Pu)	1025	10	
194	Leonard	1956	ABS+SCR(^{239}Pu)	1055	13	A15

Datapoints not added (already in DB)

Author / year	Type	Value	σ (b)	Reason
Spencer 1987	(^{235}U) ABS+SCR	690	5	Already in DB as No.166 SPENCER 1985(3) val=690.0 σ=1.0% — values match within rounding (per project decision); uncertainty discrepancy logged in A1
Spencer 1987	(^{239}Pu) ABS+SCR	1025	6	Already in DB as No.167 SPENCER 1985(3) val=1025.8 σ=1.0% — values match within rounding; uncertainty discrepancy logged in A1

Impact The new transmission datapoints sharpen ABS (and through it FIS and CA) but barely touch the SCA channel from step 4. ABS uncertainties tighten by 10–35% across the four nuclides. The largest individual tension is **Leonard 56 ABS+SCR(^{239}Pu) = 1055 b** at +2.1σ from the predicted 1027 b; this contributes ≈4.5 to the χ^2 increase of 9.5. K1 = F3ETA shifts ≤0.21% across all four nuclides relative to step 4.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	13.1683	2.559	-0.003
SCA 35	13.9536	0.922	-0.010
SCA 39	**7.4717**	5.106	**+0.173**
SCA 41	11.9949	6.665	-0.000
SCR 33	**12.0577**	5.286	**+0.542**
SCR 35	**12.0876**	6.539	**-0.716**
SCR 39	**6.6489**	17.624	**+3.085**
SCR 41	10.8359	27.592	+0.046
ABS 33	**573.7878**	0.197	**-0.112**
ABS 35	683.7585	0.185	-0.017
ABS 39	**1020.7611**	0.308	**+0.177**
ABS 41	1383.7682	1.053	+0.052
FIS 33	**528.9012**	0.248	**-0.116**
FIS 35	584.5832	0.211	-0.012
FIS 39	**749.2389**	0.304	**+0.111**
FIS 41	1018.6109	1.096	+0.037
CA 33	44.8866	1.807	-0.061
CA 35	99.1754	0.794	-0.045
CA 39	**271.5222**	0.863	**+0.359**
CA 41	365.1574	1.635	+0.093
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4915	0.184	+0.003
NUB 35	2.4304	0.169	+0.002
NUB 39	2.8824	0.191	+0.016
NUB 41	2.9444	0.208	+0.007
NUB 52	3.7668	0.129	+0.002
ETA 33	2.2965	0.199	-0.001
ETA 35	2.0779	0.190	+0.007
ETA 39	2.1157	0.264	-0.050
ETA 41	2.1674	0.390	-0.007
ALPHA 33	0.0849	1.948	+0.055
ALPHA 35	0.1697	0.875	-0.032
ALPHA 39	**0.3624**	0.937	**+0.247**
ALPHA 41	0.3585	1.573	+0.056
WGA 33	0.9988	0.120	+0.000
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	**573.1222**	0.230	**-0.112**
FA 35	669.1945	0.207	-0.017
FA 39	**1099.0534**	0.409	**+0.177**
FA 41	1445.2075	1.070	+0.052

Quantity	Value	Unc (%)	Δ vs prev (%)
FF 33	**527.0289**	0.320	**-0.116**
FF 35	571.6054	0.230	-0.012
FF 39	**791.3461**	0.410	**+0.111**
FF 41	1064.6521	1.284	+0.037
F1ETA 33	2.2911	0.307	-0.001
F1ETA 35	2.0760	0.230	+0.007
F1ETA 39	2.0754	0.466	-0.050
F1ETA 41	2.1691	0.798	-0.007
F2ETA 33	**1313.0668**	0.344	**-0.113**
F2ETA 35	1389.2322	0.273	-0.010
F2ETA 39	**2280.9637**	0.437	**+0.127**
F2ETA 41	3134.7420	1.282	+0.045
F3ETA 33	**739.9446**	0.547	**-0.114**
F3ETA 35	720.0378	0.445	-0.004
F3ETA 39	1181.9103	0.777	+0.080
F3ETA 41	1689.5344	1.755	+0.038

Step 6: All ABS+SCR → ABS+SCA; Drop Lemmel SCA-SCR; SCR Removed

Step 6 finalises the Capote chain:

- **Convert** every (ABS X)+SCR X measurement to (ABS X)+SCA X — 26 conversions covering all transmission entries from baseline plus the step-5 additions.
- **Delete** the four Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95).
- **Consequence**: with no measurement function referencing SCR and no SCA–SCR datapoint, the four SCR primary parameters become unconstrained and are removed from the fit parameter space — exactly Roberto's parenthetical *"reducing 4 fitting variables SCR"*. Decision logged in A16.

Impact relative to step 5 Differences are small. SCA channel essentially unchanged (much less than 0.02%) — the SCA additions in step 4 are the dominant constraint. ABS, FIS, CA, and the F-quantities shift by ≈0.15% as the converted transmission datapoints now correlate only with SCA, slightly relaxing the absorption fit. χ^2 rises by only 1.0 (84.2 → 85.2) despite the bulk reclassification.

Fitted quantities

Quantity	Value	Unc (%)	Δ vs prev (%)
SCA 33	13.1678	2.559	-0.004
SCA 35	13.9513	0.922	-0.016
SCA 39	7.4713	5.105	-0.005
SCA 41	11.9949	6.665	+0.000
ABS 33	**572.9047**	0.182	**-0.154**
ABS 35	**682.9942**	0.180	**-0.112**
ABS 39	1020.1612	0.301	-0.059
ABS 41	1382.9121	1.047	-0.062
FIS 33	**528.0955**	0.237	**-0.152**
FIS 35	**583.9832**	0.207	**-0.103**
FIS 39	748.8276	0.301	-0.055
FIS 41	1018.0174	1.090	-0.058
CA 33	**44.8092**	1.805	**-0.173**
CA 35	**99.0110**	0.791	**-0.166**
CA 39	271.3336	0.855	-0.069
CA 41	364.8947	1.631	-0.072
CAP 34	95.9000	2.086	+0.000
NUB 33	2.4914	0.184	-0.001
NUB 35	2.4303	0.169	-0.003
NUB 39	2.8823	0.191	-0.002
NUB 41	2.9443	0.208	-0.002
NUB 52	3.7668	0.129	-0.001
ETA 33	2.2966	0.199	+0.000
ETA 35	2.0780	0.190	+0.006
ETA 39	2.1157	0.263	+0.002
ETA 41	2.1674	0.390	+0.002
ALPHA 33	0.0849	1.948	-0.020
ALPHA 35	0.1695	0.875	-0.063
ALPHA 39	0.3623	0.934	-0.015
ALPHA 41	0.3584	1.573	-0.014
WGA 33	0.9988	0.120	+0.000

Quantity	Value	Unc (%)	Δ vs prev (%)
WGA 35	0.9787	0.092	+0.000
WGA 39	1.0767	0.269	+0.000
WGA 41	1.0444	0.191	+0.000
WGF 33	0.9965	0.201	+0.000
WGF 35	0.9778	0.092	+0.000
WGF 39	1.0562	0.275	+0.000
WGF 41	1.0452	0.670	+0.000
FA 33	**572.2401**	0.218	**-0.154**
FA 35	**668.4464**	0.202	**-0.112**
FA 39	1098.4076	0.404	-0.059
FA 41	1444.3134	1.064	-0.062
FF 33	**526.2260**	0.311	**-0.152**
FF 35	**571.0188**	0.226	**-0.103**
FF 39	790.9117	0.408	-0.055
FF 41	1064.0318	1.280	-0.058
F1ETA 33	2.2911	0.307	+0.000
F1ETA 35	2.0761	0.230	+0.006
F1ETA 39	2.0754	0.466	+0.002
F1ETA 41	2.1691	0.798	+0.002
F2ETA 33	**1311.0504**	0.336	**-0.154**
F2ETA 35	**1387.7682**	0.269	**-0.105**
F2ETA 39	2279.6646	0.435	-0.057
F2ETA 41	3132.8564	1.277	-0.060
F3ETA 33	**738.8103**	0.542	**-0.153**
F3ETA 35	719.3218	0.444	-0.099
F3ETA 39	1181.2570	0.776	-0.055
F3ETA 41	1688.5430	1.752	-0.059

Step 5 vs Step 6: Head-to-Head Comparison

This is the key comparison Roberto requested. Both candidate fits agree to better than 0.2% on every primary and derived quantity. Step 6 has 4 fewer free parameters (SCR 33, 35, 39, 41 removed) and slightly higher χ^2 (85.2 vs 84.2 — a 1.0 increase from the bulk SCR⟶SCA conversion).

Quantity	Step 5	Step 6	Δ%
SCA 33	13.1683	13.1678	-0.004
SCA 35	13.9536	13.9513	-0.016
SCA 39	7.4717	7.4713	-0.005
SCA 41	11.9949	11.9949	+0.000
SCR 33	12.0577	– (removed)	–
SCR 35	12.0876	– (removed)	–
SCR 39	6.6489	– (removed)	–
SCR 41	10.8359	– (removed)	–
ABS 33	573.7878	572.9047	**-0.154**
ABS 35	683.7585	682.9942	**-0.112**
ABS 39	1020.7611	1020.1612	-0.059
ABS 41	1383.7682	1382.9121	-0.062
FIS 33	528.9012	528.0955	**-0.152**
FIS 35	584.5832	583.9832	**-0.103**
FIS 39	749.2389	748.8276	-0.055
FIS 41	1018.6109	1018.0174	-0.058
CA 33	44.8866	44.8092	**-0.173**
CA 35	99.1754	99.0110	**-0.166**
CA 39	271.5222	271.3336	-0.069
CA 41	365.1574	364.8947	-0.072
CAP 34	95.9000	95.9000	+0.000
NUB 33	2.4915	2.4914	-0.001
NUB 35	2.4304	2.4303	-0.003
NUB 39	2.8824	2.8823	-0.002
NUB 41	2.9444	2.9443	-0.002
NUB 52	3.7668	3.7668	-0.001
ETA 33	2.2965	2.2966	+0.000
ETA 35	2.0779	2.0780	+0.006
ETA 39	2.1157	2.1157	+0.002
ETA 41	2.1674	2.1674	+0.002
ALPHA 33	0.0849	0.0849	-0.020
ALPHA 35	0.1697	0.1695	-0.063
ALPHA 39	0.3624	0.3623	-0.015
ALPHA 41	0.3585	0.3584	-0.014

Quantity	Step 5	Step 6	Δ%
WGA 33	0.9988	0.9988	+0.000
WGA 35	0.9787	0.9787	+0.000
WGA 39	1.0767	1.0767	+0.000
WGA 41	1.0444	1.0444	+0.000
WGF 33	0.9965	0.9965	+0.000
WGF 35	0.9778	0.9778	+0.000
WGF 39	1.0562	1.0562	+0.000
WGF 41	1.0452	1.0452	+0.000
FA 33	573.1222	572.2401	**-0.154**
FA 35	669.1945	668.4464	**-0.112**
FA 39	1099.0534	1098.4076	-0.059
FA 41	1445.2075	1444.3134	-0.062
FF 33	527.0289	526.2260	**-0.152**
FF 35	571.6054	571.0188	**-0.103**
FF 39	791.3461	790.9117	-0.055
FF 41	1064.6521	1064.0318	-0.058
F1ETA 33	2.2911	2.2911	+0.000
F1ETA 35	2.0760	2.0761	+0.006
F1ETA 39	2.0754	2.0754	+0.002
F1ETA 41	2.1691	2.1691	+0.002
F2ETA 33	1313.0668	1311.0504	**-0.154**
F2ETA 35	1389.2322	1387.7682	**-0.105**
F2ETA 39	2280.9637	2279.6646	-0.057
F2ETA 41	3134.7420	3132.8564	-0.060
F3ETA 33	739.9446	738.8103	**-0.153**
F3ETA 35	720.0378	719.3218	-0.099
F3ETA 39	1181.9103	1181.2570	-0.055
F3ETA 41	1689.5344	1688.5430	-0.059

Recommendation for choice between step 5 and step 6 is left to the collaborators. Step 5 retains an SCR fit constrained only by the Lemmel SCA–SCR data; step 6 abandons SCR entirely, treating all transmission data as ABS+SCA. The K1 = F3ETA differs by ≤0.15% between the two — a difference well below the percent-level uncertainties on the K1 values themselves.

K1 = F3ETA Evolution Through the Chain

This is the central derived quantity Roberto identifies as the K1 of the new TNC fit. Values across all 7 steps:

Step	F3ETA 33	F3ETA 35	F3ETA 39	F3ETA 41
step0 baseline	746.1097	721.3019	1179.5261	1692.3543
step1 lounsbury	741.9298	719.7289	1180.4095	1689.2737
step2 adamchuk	741.9189	719.5599	1180.3338	1689.1314
step3 wallner	741.8668	718.9968	1180.1470	1688.5391
step4 sca list	740.7869	720.0641	1180.9667	1688.8844
step5 transmission	739.9446	720.0378	1181.9103	1689.5344
step6 all sca	738.8103	719.3218	1181.2570	1688.5430

All four K1 values stay within ±1% of the Axton T8 baseline through the entire chain. The largest excursion is K1(^{233}U) = F3ETA(^{233}U), which drops from 746.1 (step 0) to 738.8 (step 6) — a 1.0% reduction driven primarily by the Lounsbury inclusion in step 1.

Comprehensive Dataset Table (Final Fit, Step 6)

All 124 measurements actually used in the step-6 fit (after the microscopic filter is applied). Cross-references to the Decisions appendix flag entries where a non-trivial choice was made.

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
5	LEMMEL	1982(1)	CAP 34	95.9	2.086	
21	LOUNSBURY	1970(1)	(CA 33)/FIS 33	0.0861	2.239	
22	LOUNSBURY	1970(1)	(CA 35)/FIS 35	0.1697	1.709	
23	LOUNSBURY	1970(1)	(CA 39)/FIS 39	0.3555	1.603	
54	DIVADEENAM	1984(2)	WGA 33	0.99884	0.120	
55	DIVADEENAM	1984(2)	WGA 35	0.9787	0.092	
56	DIVADEENAM	1984(2)	WGA 39	1.0767	0.269	
57	DIVADEENAM	1984(2)	WGA 41	1.0444	0.191	
58	DIVADEENAM	1984(2)	WGF 33	0.99646	0.201	

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
59	DERUYTTER	1985(3)	WGF 35	0.9778	0.092	
60	DIVADEENAM	1984(2)	WGF 39	1.0562	0.275	
61	DIVADEENAM	1984(2)	WGF 41	1.0452	0.670	
62	EGELSTAFF	1954(1)	(ABS 35)+SCA 35	724.33	3.590	
63	MELKONIAN	1953(1)	(ABS 35)+SCA 35	694.32	2.016	
64	PALEVSKY	1954(1)	(ABS 35)+SCA 35	700.32	1.428	
65	NITIKIN	1955(1)	(ABS 35)+SCA 35	710.32	2.956	
66	SIMPSON	1960(1)	(ABS 35)+SCA 35	690.32	1.391	
67	SAFFORD	1959(1)	(ABS 35)+SCA 35	696.32	0.359	
68	SAFFORD	1959(1)	(ABS 35)+SCA 35	698.32	0.730	
69	BLOCK	1960(1)	(ABS 35)+SCA 35	693.32	0.721	
70	SAPLAKOGLU	1961(1)	(ABS 35)+SCA 35	696.32	0.359	
71	GERASIMOV	1962(1)	ABS 35	670.31	2.089	
72	ZIMMERMAN	1955(1)	(ABS 39)+SCA 39	1022.5	1.271	
73	NITIKIN	1955(1)	(ABS 39)+SCA 39	1040.5	2.979	
74	BOLLINGER	1958(1)	(ABS 39)+SCA 39	1022.5	1.369	
75	PATTENDEN	A1956(1)	(ABS 39)+SCA 39	1012.5	3.003	
76	SAFFORD	1961(1)	(ABS 39)+SCA 39	1018.5	0.727	
77	MUETHER	1954(1)	(ABS 33)+SCA 33	597.27	2.344	
78	NITIKIN	1955(1)	(ABS 33)+SCA 33	577.26	3.638	A8
79	PATTENDEN	1956(1)	(ABS 33)+SCA 33	596.27	3.019	
80	GREEN	1957(1)	ABS 33	574.26	3.483	
81	SIMPSON	1960(1)	(ABS 33)+SCA 33	587.27	0.800	
82	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.77	0.990	
83	SAFFORD	1960(1)	(ABS 33)+SCA 33	585.67	0.410	A9
84	BLOCK	1960(1)	(ABS 33)+SCA 33	587.27	0.511	
85	SIMPSON	1961(1)	(ABS 41)+SCA 41	1386.6	3.606	
86	CRAIG	1964(1)	(ABS 41)+SCA 41	1380.6	2.897	
87	SMITH	1984(3)	(ABS 41)+SCA 41	1395.6	1.433	
96	DERUYTTER	1974(1)	39 FH1 39	1812.1	0.454	
97	DERUYTTER	1973(1)	34 FH1 35	1439.1	0.445	
98	DERUYTTER	1961(1)	FIS 35	589.37	1.325	
99	WHITE	1967(1)	(FIS 39)/FIS 35	1.253	1.800	
100	WHITE	1967(1)	(FIS 41)/FIS 35	1.763	2.751	
101	BERCEANO	1977(1)	34 FH1 35	1443	0.791	
102	RAFFLE	1959(1)	33 FH1 33	834.26	3.302	
103	RAFFLE	1959(1)	FIS 35	581.97	3.092	
104	RAFFLE	1959(1)	39 FH1 39	1714.8	2.843	
105	RAFFLE	1956(1)	(FIS 41)/FIS 39	1.3319	6.000	
106	MASLIN	1965(1)	FIS 35	583.71	1.449	
107	SAPLAKOGLU	1959(1)	FIS 35	593.17	2.232	
108	WATANABE	1964(1)	FIS 41	985.45	4.569	
109	FRAYSSE	1965(1)	(39 FH1 39)/34 FH1 35	1.2425	1.557	
110	BORCEA	1970(1)	(39 FH1 39)/34 FH1 35	1.2578	1.624	
111	BOLDEMAN	1977(1)	NUB 52	3.7549	0.431	
112	BOLDEMAN	1980(1)	(NUB 33)/NUB 52	0.66049	0.255	
113	BOLDEMAN	1980(1)	(NUB 35)/NUB 52	0.64318	0.255	
114	BOLDEMAN	1980(1)	(NUB 39)/NUB 52	0.76614	0.243	
115	BOLDEMAN	1980(1)	(NUB 41)/NUB 52	0.78028	0.231	
116	SPENCER	1982(1)	NUB 52	3.7831	0.221	
117	GWIN	1984(1)	(NUB 33)/NUB 52	0.65996	0.340	
118	GWIN	1984(1)	(NUB 35)/NUB 52	0.64701	0.265	
119	GWIN	1984(1)	(NUB 39)/NUB 52	0.7653	0.257	
120	GWIN	1984(1)	(NUB 41)/NUB 52	0.78423	0.290	
121	HOPKINS	1963(1)	NUB 52	3.7767	0.838	
122	HOPKINS	1963(1)	(NUB 33)/NUB 52	0.65483	0.895	
123	HOPKINS	1963(1)	(NUB 35)/NUB 52	0.64587	0.838	
124	HOPKINS	1963(1)	(NUB 39)/NUB 52	0.75047	0.996	
125	ASPLUND	1963(1)	NUB 52	3.791	1.066	
126	CONDE	1965(1)	(NUB 35)/NUB 52	0.64222	0.847	
127	WHITE	1968(1)	NUB 52	3.8194	1.033	
128	AXTON	A1985(3)	NUB 52	3.7547	0.300	
129	COLV/AXT	1966(1)	NUB 52	3.7299	0.806	
130	COLV/ULL	1965(1)	NUB 52	3.7405	0.438	
131	COLVIN	1965(1)	(NUB 33)/NUB 35	1.0204	0.587	
132	COLVIN	1965(1)	(NUB 39)/NUB 35	1.1835	0.675	
133	COLVIN	1965(1)	(NUB 41)/NUB 35	1.2102	0.905	
134	COLVIN	1965(1)	(NUB 35)/NUB 52	0.64334	0.472	
135	MATHER	1964(1)	(NUB 33)/NUB 52	0.67093	1.235	
136	MATHER	1964(1)	(NUB 35)/NUB 52	0.64145	0.537	
137	MATHER	1964(1)	(NUB 39)/NUB 52	0.77307	1.122	
138	ALEK-ROV	1981(1)	NUB 52	3.7618	0.483	
139	SMITH	1984(1)	NUB 52	3.7678	0.303	

No.	Author	Reference	Quantity	Value	σ (%)	Appendix
140	EDWARDS	1982(1)	NUB 52	3.7641	0.711	
141	BOZ-NESH	1977(1)	NUB 52	3.7475	0.580	
142	DEVOLPI	1972(1)	NUB 52	3.7507	0.463	
143	ZHANG	1981(1)	NUB 52	3.7534	0.490	
144	SPIEGEL	1981(1)	NUB 52	3.7828	0.759	
145	SMITH	1984(1)	ETA 33	2.2972	0.350	
146	SMITH	1984(1)	ETA 35	2.0842	0.386	
147	SMITH	1984(1)	ETA 39	2.112	0.380	
148	SMITH	1984(1)	ETA 41	2.169	0.419	
149	MACKLIN	1960(1)	ETA 33	2.3121	0.375	
150	MACKLIN	1960(1)	ETA 35	2.0757	0.382	
151	MACKLIN	1962(1)	ETA 39	2.1163	0.851	
165	LLOYD	1982(1)	(ETA 39)/FIS 39	0.002782	2.072	
166	SPENCER	1985(3)	(ABS 35)+SCA 35	690	1.000	A1
167	SPENCER	1985(3)	(ABS 39)+SCA 39	1025.8	1.000	A1
168	ADAMCHUK	At. En. (R), 65, 434 (1988)	(CA 35)/FIS 35	0.169	2.100	
169	WALLNER	PRL 112, 192501 (2014)	CA 35	99.5	2.000	
170	WALLNER	PRL 112, 192501 (2014)	CA 35	101.5	2.000	
171	ATOYI	Acta Cryst. 20, 587 (1966)	SCA 35	12.3	4.065	A2
172	WILLIS	PRS A274, 122 (1963)	SCA 35	12.8	7.812	A3
173	KAISER	Physica B 136, 134 (1986)	SCA 35	14	1.429	
174	ARIF	PR A35, 2810 (1987)	SCA 35	14	1.429	
175	CEULEMANS	Divadeenam compilation 1970	SCA 35	14.6	3.425	A4
176	BAYVAS	Divadeenam compilation 1980	SCA 35	14.3	10.489	
177	FOOTE	Divadeenam compilation 1958	SCA 35	15	6.667	
178	OLEKSA	Phys Rev 109, 1645 (1958)	SCA 33	13.4	3.731	A5
179	GREEN	JNM 51, 281 (1974)	SCA 33	12.6	5.556	A6
180	VERTEBNYI	Divadeenam compilation 1974	SCA 33	13.2	4.545	
181	GREEN	JNM 34, 281 (1970)	SCA 39	7.7	7.792	
182	ATOYI	Acta Cryst. 20, 587 (1966)	SCA 39	7.3	9.589	
183	ROOF	Acta Cryst. 15, 351 (1962)	SCA 39	7.3	9.589	
184	PRONYAEV	Internal analysis 2026-04-17	SCA 41	12	6.667	A7
185	ANTONOV	Capote 2026-05-04 list	(ABS 35)+SCA 35	695	2.158	A10
186	GERASIMOV	Capote 2026-05-04 list	(ABS 35)+SCA 35	685	1.314	A11
187	LEONARD	Capote 2026-05-04 list (1954)	(ABS 35)+SCA 35	702	1.994	A12
188	PSCHENICHNYI	Capote 2026-05-04 list (1978)	(ABS 33)+SCA 33	587.9	0.510	A13
189	HARVEY-1	Capote 2026-05-04 list (1979)	(ABS 33)+SCA 33	585	0.239	A14
190	HARVEY-2	Capote 2026-05-04 list (1979)	(ABS 33)+SCA 33	592	1.014	A14
191	SCHWARTZ	Capote 2026-05-04 list (1958)	(ABS 41)+SCA 41	1410	5.674	
192	ANDERSON	Capote 2026-05-04 list (1945)	(ABS 39)+SCA 39	1045	2.392	
193	PALEVSKY	Capote 2026-05-04 list (1955)	(ABS 39)+SCA 39	1025	0.976	
194	LEONARD	Capote 2026-05-04 list (1956)	(ABS 39)+SCA 39	1055	1.232	A15

A. Decisions and Contentious Points

Every choice that involved interpretation, departure from a literal reading of Roberto's emails, or risk of double-counting is recorded here so collaborators can scrutinise it.

A1. Spencer 1987 vs. existing No.166/167 — values used as-is

Roberto's "Revision/additions" email lists Spencer 1987 ABS+SCR(^{235}U) = 690(5) and Spencer 1987 ABS+SCR(^{239}Pu) = 1025(6). The baseline already contains:

- No.166 SPENCER 1985(3) ABS+SCR(^{235}U) = 690.0, σ = 1.0% (= 6.9 b), with a single systematic component `alpha = 0.707`.
- No.167 SPENCER 1985(3) ABS+SCR(^{239}Pu) = 1025.8, σ = 1.0% (= 10.26 b), same systematic.

The values match within rounding (690 vs 690.0; 1025 vs 1025.8 — likely the same measurement, presented at a 1985 conference and published 1987). Per project decision, the database entries are retained as-is rather than being re-overwritten with Roberto's values. **Note**: the uncertainties differ noticeably (DB: 1.0%; Roberto: 0.72% / 0.59%); the more conservative DB uncertainty is what enters the fit. Step-6 conversion turns these into ABS+SCA(^{235}U) and ABS+SCA(^{239}Pu) like all other ABS+SCR entries.

A2. Atoyi 1966 ABS+SCR tag treated as SCA

In the SEARS column of Capote's 2026-04-30 inlined compilation, the Atoyi 1966 row is annotated "SCR". The same row also gives Sears Sel = 12.3(5) b. Per project decision, the entire compilation is treated as providing SCA values regardless of the per-row "SCR/SCA" tag, because (a) Sears' formalism delivers the bound-atom S_el (= SCA) by construction, and (b) the per-row tags in Roberto's table show inconsistencies that are most plausibly explained as paste artifacts from a heterogeneous source table.

A3. Willis 1963 — EXFOR value used

The Sears compilation gives Willis 1963 = 12.1(5) coh + 0.2(2) inc ⟶ S_el 12.3(5). Roberto annotates "replace with EXFOR value" and the Divadeenam section gives "WILLIS 1963 12.8(1.0) = EXFOR". The EXFOR value 12.8(1.0) is used in step 4.

A4. Ceulemans 1970 — V-51 renormalisation used as-is

Roberto's table notes "EXFOR 14.3(5) rel V-51 (1.02)" — the 14.6(5) value used in step 4 is 14.3 × 1.02, the 1.02 factor being a renormalisation against vanadium-51 standard. Different evaluations of V-51 thermal scattering vary by ~1–3% (Hughes-Schwartz 1958: ~5.0 b; Sears 1992 / ENDF/B-VIII: 5.08 b; ratio ≈ 1.016). Re-renormalising against current standards would change Ceulemans by < 1% (≈ 0.05–0.15 b), well under the quoted σ = 0.5 b — no practical impact on the fit. **Used as-is per project decision.**

A5. Oleksa 1960 — quoted at 0.1 eV, used as-is

The EXFOR replacement Oleksa 13.4(5) is quoted "at 0.1 eV", not at the thermal point 0.0253 eV. Two candidate corrections were estimated:

- **Bound⟶free transition**: σ_free / σ_bound = $(A/(A+1))^2$ = 0.9914 for A=233, i.e. <0.3% effect at 0.1 eV (still in bound regime); ≤ 0.04 b.
- **Resonance contribution**: U-233 has no large resonance below 1 eV. Tail contributions from negative resonances and the 1.45 eV resonance at 0.1 eV are at most ≈ 0.05 b.

Combined |Δσ| < 0.1 b, well below quoted σ = 0.5 b. Used as-is per project decision.

A6. Green 1974 ABS+SCR tag treated as SCA

Same situation as A2: the Divadeenam-section row for Green 1974 carries a "SCR" annotation but is added as SCA(^{233}U) per the global decision to treat all compilation rows as SCA.

A7. Pu-241 SCA — Pronyaev internal analysis

No direct experimental SCA measurement exists for ^{241}Pu. Pronyaev's 2026-04-17 internal analysis derives SCA(^{241}Pu) = 12.0(0.8) b from the potential scattering radius R_pot = 9.54 fm (consistent across ENDF/B-VIII.1, JEFF-4.0 and JENDL-5; $4\pi R^2$ = 11.44 b plus the contribution of negative resonances in the dominant ENDF/B-VIII.1 evaluation). This estimate is encoded as a direct SCA(^{241}Pu) "measurement" replacing the deleted Lemmel No.94 (12.006 b, σ = 21.6%). The replacement reduces the SCA(^{241}Pu) uncertainty from 21.6% (Lemmel) to 6.7% (Pronyaev).

A8. Nikitin 1955 spelling — DB uses "NITIKIN"

Roberto writes "Nikitin" (correct transliteration of the Russian Никитин); the database has the typo "NITIKIN" inherited from Axton 1986 throughout (Nos. 65, 73, 78). The DB spelling is preserved for stable cross-referencing.

A9. Safford 1960 vs 1959 ambiguity

Roberto's "Revision/additions" email instructs *"Change Safford 1959 ABS+SCR to ABS+SCA"* for U-233. The database has no Safford 1959 U-233 entry; Safford's U-233 transmission measurements are dated 1960 (Nos. 82 ABS+SCA(^{233}U) = 585.77 already, 83 ABS+SCR(^{233}U) = 585.67). The 1959 ⟶ 1960 mismatch is treated as a typo and No.83 is reclassified per Roberto's evident intent. Worth verifying with Roberto if there is an actual 1959 paper we are missing.

A10. Antonov 1986 ABS+SCA tag — non-default for transmission

Roberto tags Antonov 1986 ^{235}U as ABS+SCA, which is unusual for a transmission measurement (the dominant convention is ABS+SCR for rolled-metal samples). The tag is honoured under the assumption that Antonov used a non-metallic sample (UO_2 powder?) or applied a proper extinction correction. Verify with Roberto / read the original paper.

A11. Gerasimov 1962 — likely separate from existing No.71

A GERASIMOV 1962 entry exists in the baseline as **No.71 GERASIMOV 1962(1) ABS 35 = 670.31 b, σ = 2.09%** (bare ABS, no scatter component). Roberto's "Gerasimov 62 ABS+SCR 685(9)" has a different value (685 vs 670.3) and a different quantity (ABS+SCR vs ABS). Per project decision, this is added as a separate measurement (No.186) rather than overwriting No.71. Worth verifying with Roberto if this is in fact the same measurement re-analysed.

A12. Leonard 1954 — type defaulted to ABS+SCR

Roberto's email gives "Leonard 54 702(14)" with no SCR/SCA tag. Defaulted to ABS+SCR per the convention for the era (rolled-metal samples) and pending paper lookup. This decision is **moot for the final fit** because step 6 converts all ABS+SCR to ABS+SCA in any case.

A13. Pschenichnyi 1978 uncertainty — typo fix 0.3 → 3.0

Roberto's email reads *"Pschenichnyi 78 ABS+SCR 587.9(3)"*, which would be σ = 0.3 b on 587.9 b ≈ 0.05% — implausibly small for a transmission measurement. **Treated as a typo and used as σ = 3.0 b** (≈ 0.51%). This decision is conservative; if Roberto truly intended 0.3 b, Pschenichnyi would dominate the U-233 absorption fit by orders of magnitude.

A14. Harvey-1 / Harvey-2 — kept uncorrelated; correlation alternative noted

The two Harvey 1979 ABS+SCR(^{233}U) entries come from the same author, same year, and likely the same ORELA campaign — physical correlation through sample mass / normalisation is plausible. Per project decision, the two are entered with empty `systematic_uncertainties` and treated as independent, matching the convention for nearly all other Axton transmission entries.

Alternative considered (not adopted): introduce a shared code `harvey79` with a 0.20% common component:

```
No.189 Harvey-1: σ_total = 0.24%; sys = {harvey79: 0.20}
No.190 Harvey-2: σ_total = 1.01%; sys = {harvey79: 0.20}
```

This would marginally tighten the U-233 ABS uncertainty. Worth revisiting if Roberto wants tighter modelling of correlated systematics.

A15. Leonard 1956 ABS+SCR(^{239}Pu) — largest tension in step 5

Leonard 1956 ABS+SCR(^{239}Pu) = 1055 b lies +2.1σ above the fit-predicted 1027.4 b. This contributes ≈4.5 of the 9.5 χ^2 increase between step 4 and step 5. The measurement is retained as-is; flagged here as the dominant transmission tension in the chain.

A16. SCR removed from parameter space in step 6

Roberto's step 6 instructs *"CONVERT ALL SCR in ABS+SCR to ABS+SCA, delete Lemmel SCA-SCR (reducing 4 fitting variables SCR)"*. The literal application:

- Convert every (ABS X)+SCR X → (ABS X)+SCA X (26 conversions).
- Delete the four Lemmel SCA–SCR datapoints (Nos. 89, 91, 93, 95).
- **No measurement function or datapoint references SCR after these changes**, so the four SCR primary parameters become unconstrained.
- They are therefore removed from the fit parameter space, consistent with the existing handling of unconstrained parameters (CA 40, CA 42, GC116, GA116). This matches Roberto's parenthetical "reducing 4 fitting variables SCR" exactly.

A17. Microscopic filter and unconstrained parameters

All seven benchmarks apply the same microscopic filter that reproduces Axton Table 8: exclude measurements whose `measured_function` references any Maxwellian-derived quantity (FA, FF, CAP, FFH, F1ETA, F2ETA, F3ETA, FLEM, F1BIG, F1CAB–F5CAB, GC116, GA116) — except CAP 34 (No.5), which is retained because it ≈ σ_γ for non-fissile U-234. Lounsbury alpha (Nos. 21–23) is excluded only in step 0; it is included in steps 1–6. Unconstrained parameters CA 40, CA 42, GC116/40/41/42, GA116/39/41 are removed from the fit parameter space throughout. Step 6 additionally removes SCR 33/35/39/41 (see A16).

A18. Empty systematic uncertainties for step-5 transmission additions

All step-5 transmission additions are entered with empty `systematic_uncertainties` (single uncorrelated total uncertainty), matching the convention used for 26 of the 28 baseline transmission entries. Spencer 1985 (Nos. 166/167) is the existing exception with `alpha = 0.707` representing the α-correction Spencer used internally. The Harvey-pair correlation alternative is the only other systematic structure considered (see A14).

B. Files Produced by This Chain

All on branch `tnc_update_capote_20260504`, building on commit `575fac0` (the Axton 1986 reproduction state of master).

JSON databases (one per step where data changes)

- `axton1986/axton1986_data.json` — baseline (steps 0, 1 use this directly)
- `axton1986/axton1986_data_capote_step2_adamchuk.json` — 168 measurements
- `axton1986/axton1986_data_capote_step3_wallner.json` — 170 measurements
- `axton1986/axton1986_data_capote_step4_sca_list.json` — 180 measurements
- `axton1986/axton1986_data_capote_step5_transmission.json` — 190 measurements
- `axton1986/axton1986_data_capote_step6_all_sca.json` — 186 measurements (final)

Benchmarks

- `evaluation/benchmark_capote_step0_baseline.py`
- `evaluation/benchmark_capote_step1_lounsbury.py`
- `evaluation/benchmark_capote_step2_adamchuk.py`
- `evaluation/benchmark_capote_step3_wallner.py`
- `evaluation/benchmark_capote_step4_sca_list.py`
- `evaluation/benchmark_capote_step5_transmission.py`
- `evaluation/benchmark_capote_step6_all_sca.py`
- `evaluation/benchmark_capote_summary.py` — runs all 7 in sequence and dumps `capote_chain_summary.json`

Generated artefacts

- `evaluation/capote_chain_summary.json` — machine-readable consolidation of all step results
- `evaluation/report_capote_fit.md` — this report
- `evaluation/build_capote_report.py` — the script that generated this report from the summary JSON